

Research Statement

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1 Introduction

The focus of my research is on developing and analyzing efficient and robust numerical methods for the simulation of complex real-world systems and their uncertainty quantification (UQ). My research experience spans computational mathematics and computer science, and their applications to real-world multi-physics problems such as biomolecular electrostatics, electrical activity of the heart, chemical reactions, magnetohydrodynamics and oceanic flows.

The main contributions are in the areas of **adjoint based *a posteriori* error estimation**, **least-squares finite element methods**, and **reduced order modeling**. My recent research has focused on developing novel numerical methods and mathematical analyses which have significantly advanced the frontiers in these fields and have had deep impact in computational science. My research highlights include:

- Error estimation for the Poisson-Boltzmann equation which takes into account the regularity of the adjoint solution;
- A robust reduced order method which considers the error between the low order solution and the true solution;
- *A posteriori* error analysis for the time-parallel Parareal algorithm;
- An error analysis for the first time to a threshold event (e.g. when a tsunami wave reaches a certain height);
- Uncertainty quantification for problems posed on stochastic domains.

By tackling directly the complex issues associated with these solution techniques and application areas, my research has produced new and reliable error estimates for these widely used methods and important physical phenomenon. The estimates derived in this research provide guidance for choosing values of numerical parameters needed to obtain a desired accuracy. They also form a cornerstone of reliable use of uncertainty quantification techniques in science and engineering.

2 *A posteriori* error estimation

My research has tackled numerous challenges in forming robust and accurate error estimates for complex PDES:

- Developing an appropriate adjoint operator for iterative schemes e.g. iteration between sub-domains, iteration between time scales, iteration between multi-rate components of a PDE etc [5, 10, 17, 18];
- Development of error estimates accounting for the regularity and well-posedness of the adjoint problem [6, 7];
- Mathematical treatment of non-computable terms in the error estimate by showing they are higher order relative to the other terms and hence may be ignored in the limit of discretization. This often involves delicate analysis [17, 4];
- Estimation of other non-computable terms in the estimate that are not higher order. This is achieved by defining auxiliary adjoint problems with Quantities of Interest that accurately quantify the contributions due to these terms [10].

Some of my recent novel analyses for these and other challenges are described below in more detail.

2.1 Background

The use of complex solution techniques for nonlinear differential equations typically means the numerical solution has large error. Hence, the reliable use of simulations in science and engineering necessitates quantifying the error in the numerical solution. Tackling this goal of *a posteriori* error analysis has formed the corner stone of my research activities at UNM.

The main mathematical tools used in *a posteriori* error analysis are adjoint problems and computable residuals. Consider a differential equation,

$$Lu = g(\mathbf{x}, t), \quad (1)$$

where L denotes a linear differential operator. We are interested in computing the error in a Quantity of Interest (QoI) expressed as a linear functional (u, ψ) where $(\cdot, \cdot)$ denotes the (space-time) L_2 inner product and ψ is chosen to yield the desired information. Let L^* be the adjoint operator to L and consider the adjoint problem

$$L^*\phi = \psi. \quad (2)$$

This yields the error estimate

$$\text{Error in the QoI} = (u - U, \psi) = (u - U, L^*\phi) = (L(u - U), \phi) = (g - LU, \phi). \quad (3)$$

The extension of this approach to nonlinear differential equations is achieved by linearization. However, additional complications arise due to considerations of the regularity and well-posedness of the adjoint solution. In addition, most solution methods for nonlinear multi-physics systems involve sophisticated multi-discretization techniques such as domain decomposition, implicit-explicit time integration schemes as well as the use of iteration in various guises. Consequently, the adjoint of the discrete solution operator becomes much more complicated to define. My research has addressed these challenges as I detail below.

2.2 Error Estimation for a novel Quantity-of-Interest

In *Error estimation and uncertainty quantification for first time to a threshold value*, I develop an *a posteriori* analysis for a quantity of interest that cannot be represented as a linear functional (nor can it be easily linearized). This is the first known analysis of such a QoI that considers the time at which the solution crosses a threshold value for the first time. This QoI is of great importance in science and engineering applications such as when a tsunami wave reaches a certain height or when the concentration of an ion species exceeds a certain limit.

First, we show an *a priori* result for the convergence of this “non-standard” QoI. Then, we carry an *a posteriori* analysis to quantify the error in the numerically computed value of this non-standard QoI. The resulting error estimate may be presented at a high level as [4],

$$\text{Error in non-standard QoI} = \frac{\text{Error in a standard QoI} + \text{H.O.T}}{\text{Error in another standard QoI} + \text{H.O.T}}. \quad (4)$$

Here the standard QoIs refer to bounded linear functionals, the error in which is computed using traditional adjoint based analysis. The H.O.T refers to higher order terms which are small in magnitude relative to the other terms in the error estimate, and hence may be dropped. We also compare the above estimate to techniques based on standard nonlinear solvers. We demonstrate that the estimate (4) is computationally more efficient than traditional methods as it requires approximately five times fewer adjoint solutions while having the same accuracy.

2.3 *A posteriori* Error Analysis for complex PDEs

The Poisson-Boltzmann Equation

My work *A posteriori analysis and efficient refinement strategies for the Poisson– Boltzmann equation* develops a well-posed adjoint problem to the nonlinear Poisson-Boltzmann Equation (PBE) and develops novel adaptive algorithms to arrive at accurate solutions. The naive formulation of the adjoint problem leads to an ill-posed system due to the loss of regularity of the solution. This research complements my earlier work on developing and analyzing numerical methods for a variety of problems in biomolecular electrostatics that have had a significant impact on the field as evidenced by citations of the articles [1, 15, 17, 14, 13, 2] by various researchers.

The PBE is a continuum model for biomolecular electrostatics and describes several important biological systems. The PBE in its classical form is not amenable to such analysis, and is often split into three terms representing a singular component, a regular component and a harmonic component. Due to the complex, nonlinear nature of the PBE, the numerical solution is challenging and often has large error. In my work, I quantify the error in an important biological QoI: the solvation free energy. This is the energy needed to transfer a charged molecule from vacuum to an ionized solution. The adjoint operators to the regular and harmonic components are defined by accounting for the coupled nature of the three term split PBE in addition to the issues arising due to the regularity of the solution (the singular component has an analytical formula and hence does not have any numerical error). The resulting error estimates are shown to be accurate. The error is partitioned in such a way that specific sources of error are identified and addressed. The main theorem of this research is summarized as:

Theorem 1 *The error in the solvation free energy computed from the finite element solution of the nonlinear PBE is the sum of following contributions:*

- *Finite element discretization error for the regular component;*
- *Finite element discretization error for the harmonic component;*
- *Interface discretization error; and*
- *Error due to approximation of boundary conditions.*

Moreover, novel refinement schemes, called Uniform Contribution Refinement and Adaptive Contribution Refinement, utilize the information about the sources of error to arrive at accurate computed values of the QoI [7].

Magnetohydrodynamics

In recent work *A posteriori error analysis for stationary incompressible magnetohydrodynamics* we develop robust error estimates for the magnetohydrodynamics (MHD) system of equations. In particular, the definition of the adjoint problem was not immediate given that the multiphysics MHD system is a rectangular system (with more equations than unknowns). Thus, we construct the adjoint based on the exact penalty weak form of the MHD system. Then, we employ the setting of saddle point problems to show that the appropriate weak forms of the adjoint problem satisfy the boundedness, coercivity on a certain subspace and inf-sup conditions, and hence the adjoint problem is well posed. The error estimate decomposes the error due to the different physics and provides guidance for discretization choices [6]

2.4 Analysis of parallel-in-time methods

In *A posteriori error analysis of two-stage computation methods with application to efficient discretization and the Parareal algorithm* we construct accurate yet computationally inexpensive error estimates for the coarse and fine scale solutions of the Parareal algorithm. This research is critical for the reliable use of time-parallel Parareal algorithm on exascale computers. The Parareal method is an iterative method that involves two scales; a computationally inexpensive but serial coarse scale and a computationally expensive but parallel fine scale. Information is exchanged between the two scales at certain synchronization points.

The adjoint based analysis we devised in [10] constructs accurate error estimates for both the fine scale and coarse scale solutions. The key in constructing the estimate was to interpret the information exchange between the two scales as not modifying the solution at the synchronization points, but as changing the “initial conditions” for the numerical solution at those points. This led to a variational version of Parareal algorithm that was amenable to adjoint based error estimation. We showed that the variational version of the algorithm was equivalent to the traditional version.

While adjoint based analysis could be carried out on the variational version, solving the adjoint problem in serial on the fine scale would be computationally impractical. Thus, my work devised a adjoint solution strategy that mimicked the Parareal solution strategy: a coarse scale serial solve and a fine scale parallel solve. However, this meant that the resulting error estimate had non-computable terms involving the true solution. In this case, we defined additional auxiliary adjoint problems to account for these terms. The auxiliary adjoint problems were posed on the coarse scale and hence were inexpensive to compute. The final error estimate decomposed the error as

- Iteration error contribution;
- Discretization error contribution;
- Coarse scale error contribution; and
- Auxiliary error contribution.

Information about each of these error contributions may be used by practitioners to improve the solution accuracy. This research enables efficient yet reliable solution of real-world problems on the future’s highly parallel machines and which takes into account all sources of numerical errors.

2.5 Error analysis for IMEX schemes

In a series of papers, *A posteriori error estimation for multi-stage Runge–Kutta IMEX schemes*, *A posteriori analysis of an IMEX entropy-viscosity formulation for hyperbolic conservation laws with dissipation* and *A posteriori error analysis of IMEX multi-step time integration methods for advection-diffusion-reaction equations*, we develop error estimates for implicit-explicit time integration schemes. The adjoint based analysis relied on devising certain projection operators and quadrature rules [12, 9, 16].

2.6 Error analysis for Domain Decomposition

In recent work *A posteriori error analysis for Schwarz overlapping domain decomposition methods* we develop error estimates for additive and multiplicative Schwarz overlapping domain decomposition methods. A key ingredient of the analysis was to recognize that the Schwarz iteration corresponded to effectively defining a *modified* differential operator, distinct from the original PDE operator. The adjoint to this modified differential operator was itself iterative in nature. The analysis derived estimates for both the iteration and discretization errors in Schwarz methods [5].

3 Uncertainty quantification for problems posed on stochastic domains

The focus of this research is on uncertainty quantification and efficient solution of boundary value problems for elliptic partial differential equations (PDEs) posed on domains with stochastically perturbed boundaries. The problem of stochastic boundaries occurs for a variety of reasons, e.g. from physical stresses, manufacturing deficiencies, and uncertainty in measurements of a fixed geometry. Two interesting issues arise in such problems:

- Since the geometric properties of the domain has a strong effect on solution behavior and smoothness, significant variation in solution behavior for different realizations of the domain is to be expected. Correspondingly, significant variation in the error arising from discretization and sampling is also to be expected;
- Each realization of a domain nominally requires construction of a new discretization mesh, at a significant computational cost. Hence, solving such problems using a Monte Carlo approach is computationally intensive.

My research deals with these issues using two ideas. First, a family of simple, locally determined, well-behaved transformations is constructed that map a given elliptic problem posed on a family of stochastic domains to an elliptic problem with stochastic coefficients posed on a fixed reference domain. Second, uncertainty quantification is carried out by deriving *a posteriori* error estimates for the approximate cumulative distribution function for a Quantity-of-Interest (QoI) computed from a numerical solution that take into account all sources of deterministic and stochastic errors. The estimate is sufficiently detailed to enable balancing of computational work, e.g. mesh resolution versus sample size, to achieve a desired accuracy. This is illustrated in Figure 1 where the error bound in computing the the cumulative distribution function along with the contributions to the error from discretization and sampling are shown. Lastly, computational efficiency is addressed by devising an efficient adaptive strategy that leads to a mesh that produces acceptable accuracy for all realizations of the problem [8].

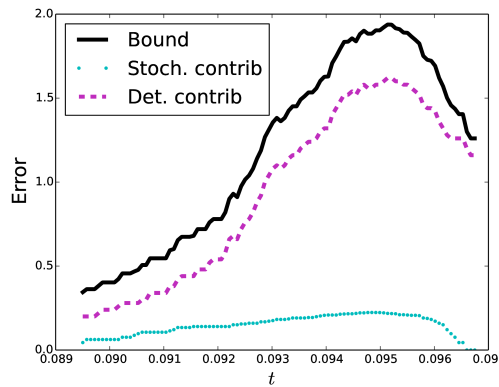


Figure 1: Error in the CDF for a convection-diffusion problem.

4 Reduced Basis Methods for Least-Squares Finite Element Methods

In this research, a greedy reduced basis method is constructed such that the next basis vector is chosen based on the error between the reduced basis (RB) solution and the true solution. This is a significant step forward in existing algorithms for RB construction, which typically chose the next basis vector based on error between the RB solution and the finite element (FE) solution. A key and necessary ingredient in constructing such a novel RB algorithm is the use of least-squares finite element methods, which are a class of finite element methods based on theoretically sound optimization framework.

Reduced order modeling techniques reduce the complexity of the numerical solve for a PDE, are thus essential for applications which either require multiple solutions for different parameters or have realtime running requirements. Reduced basis methods are one such reduced order modeling technique which employ a spectral basis to express the discrete solution. Reduced basis (RB) methods have an expensive offline stage for the computation of the spectral basis and a fast online stage for computation of the numerical solution. In earlier work, I devised a hierarchical method for constructing the RB based on proper orthogonal decomposition [3]. In my recent research, I focus on the “greedy” algorithm for construction of the RB basis.

The greedy algorithm constructs the reduced basis space by adding one basis vector at every step. Prior to my work, this new basis vector was chosen as the solution, computed from the existing basis, which exhibits the largest *a posteriori* error bound with respect to the FE error. If the FE solution itself has a large error, then greedy algorithms are of limited value. My research formed a novel error bound which compared the RB solution with the true solution. In addition, the traditional error bounds used in RB methods may over-estimate the error by orders of magnitude. In my research, by using a novel splitting of the error, a sharp and rigorous error bound is formed [11]. The sharpness of the error bound is illustrated in the Figure 2.

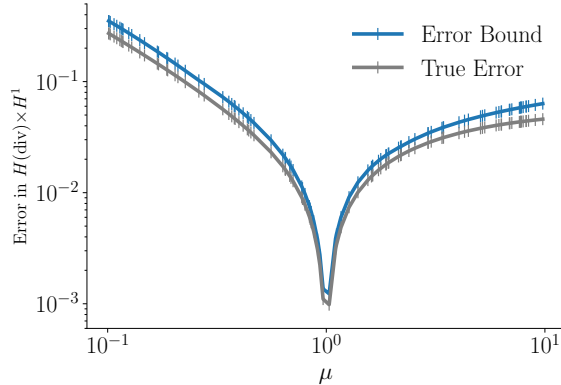


Figure 2: The error between the RB solution and the reference solution, along with the novel RB error bound over the testing set of parameter values of μ .

5 Impact and Outlook

My research has garnered citations from a number of areas, and applications are using and extending my work in error estimation and design of numerical methods. The impact of my contributions is also reflected in the external funding I have received from both National Science Foundation, Sandia National Laboratories and Lawrence Livermore National Laboratory:

- Title: “Collaborative Research: A posteriori Error Analysis for Complex Models with Applications to Efficient Numerical Solution and Uncertainty Quantification”. Principal investigator: Jehanzeb H. Chaudhry. Funding organization: National Science Foundation. NSF-DMS-1720402 (Computational Mathematics). August 2017 – July 2020. Amount awarded \$100,160.
- Title: “A Parallel-in-Time Multigrid Methods for Hyperbolic Problems, with a Focus on the Shallow Water Equations”. Co-PI sub-contract through Lawrence Livermore National Laboratory, Livermore, CA. June 2017. Total Award Amount \$69,971.
- Title: “Adjoint error estimation for Entropy viscosity formulations for Euler Equations”. Sub-contract through Sandia National Laboratories, Albuquerque, NM. June 2017. Amount \$17,500.
- Title: “Adjoint based analysis and verification for Implicit-Explicit Schemes for Shock-Hydrodynamics”. Principal investigator: Jehanzeb H. Chaudhry (with John Shadid and Tim Wildey at Sandia National Laboratories). Funding organization: Sandia National Laboratories: Laboratory Directed Research and Development (LDRD) Funding under Academic Alliance Program. February 2016 – September 2016. Amount awarded: \$50,000.

I involve graduate students in this research and fosters an environment in which they can develop and achieve their research goals. Currently, I am advising four graduate students; two for the master’s and two for PhD. My PhD students have both passed their comprehensive exams and are on track to defend their thesis either next year or the soon after. Both the master’s students are defending their thesis in the upcoming summer (2020).

Moving forward, I am excited about several important challenges in scientific computing related to and exposed by my research. In particular, a long term goal is to enable the reliable use of UQ techniques in science and engineering. As such, my goal is form an *a posteriori* error analysis for the state of the art UQ methods like the Multilevel Monte Carlo (MLMC) and Multifidelity Monte Carlo (MFMC) methods. These methods are generalizations of the classical Monte Carlo approach which greatly reduce the computational cost.

I envision research in uncertainty quantification and *a posteriori* error analysis using adjoint based methods for complex multi-scale and multi-physics problems, analysis of multilevel Monte Carlo methods, analysis of parallel-in-time schemes and application of these ideas to problems in science and engineering.

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